

A Convex Partial Answer to Smoothing Value Functions with Upper-Level-Dependent Lower Constraints

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Abstract

Alcantara and Takeda ask whether value-function smoothing frameworks for simple bilevel problems can be extended to lower-level feasible sets that depend on the upper-level variable. In particular, they ask whether a quadratic-regularization/penalization approximation is a smoothing family and whether it is gradient consistent. This note gives a rigorous partial answer, not a full resolution of that general question. We treat the convex parametric model

$$v(x) = \min\{g(x, y) : y \in C, h_i(x, y) \leq 0, i = 1, \dots, r\},$$

where C is compact convex, g is finite closed convex, the constraint functions h_i are C^1 and jointly convex, and a local uniform Slater condition holds. We construct

$$v_\mu(x) = \min_{y \in C} \left\{ e_\mu g(x, y) + \frac{1}{2\mu} \sum_{i=1}^r [h_i(x, y)]_+^2 + \frac{\mu}{2} \|y\|^2 \right\},$$

where $e_\mu g$ is the Moreau envelope. We prove that each v_μ is C^1 , that $v_\mu(x) \rightarrow v(\bar{x})$ whenever $(x, \mu) \rightarrow (\bar{x}, 0)$, and that

$$\emptyset \neq \limsup_{x \rightarrow \bar{x}, \mu \downarrow 0} \nabla v_\mu(x) \subseteq \partial_C v(\bar{x}).$$

In the smooth convex case the Moreau envelope may be omitted, yielding the pure quadratic-penalty-plus-regularization approximation. The result is complete within the stated convex/Slater setting, but it does not settle the broader nonconvex or merely Lipschitz setting of the open problem.

1 Introduction and scope

For a fixed lower-level feasible set Y , the value-function approach to bilevel optimization studies

$$v(x) = \min_{y \in Y} g(x, y),$$

and seeks smooth approximations v_μ with both approximation and gradient-consistency properties. Alcantara and Takeda developed such frameworks for nonsmooth simple bilevel problems and, in their concluding discussion, identified the extension to lower-level sets depending on x as an open direction; they note in particular that a quadratic-regularization/penalization approach had been proposed, but that smoothing-family and gradient-consistency properties remained to be established in that setting [1, 2].

This note proves such properties under a restrictive but natural convex parametric-programming hypothesis. The lower-level feasible set has the inequality form

$$Y(x) = \{y \in C : h_i(x, y) \leq 0, i = 1, \dots, r\},$$

where C is compact convex, the constraint functions are jointly convex and smooth, and a local uniform Slater point is available. The statement should therefore be read as *partial progress*: it solves the smoothing and gradient-consistency question for this convex subclass, but it does not solve the full problem for arbitrary x -dependent lower-level sets or nonconvex lower-level programs.

The proof has three ingredients. First, the quadratic term $\frac{\mu}{2} \|y\|^2$ makes the penalized lower problem single-valued, so Danskin's theorem gives differentiability of v_μ . Second, the local Slater point repairs feasible points as x moves, proving convergence of values. Third, the same Slater point bounds the exterior-penalty multipliers; any cluster gradient is then identified with a Lagrangian subgradient of the limiting convex parametric problem and hence belongs to the Clarke subdifferential of v .

2 Notation and standing assumptions

All spaces are finite dimensional and carry the Euclidean norm. For a closed convex set $C \subset \mathbb{R}^m$, $N_C(y)$ denotes the convex normal cone

$$N_C(y) := \{\nu \in \mathbb{R}^m : \langle \nu, z - y \rangle \leq 0 \text{ for all } z \in C\}.$$

For a locally Lipschitz function ϕ , $\partial_C \phi(\bar{x})$ denotes the Clarke subdifferential and $\widehat{\partial} \phi(\bar{x})$ the Frechet subdifferential. For a proper closed convex function, $\partial \phi$ denotes the convex subdifferential.

Let $\Omega \subset \mathbb{R}^n$ be open. Given a nonempty set $C \subset \mathbb{R}^m$ and functions h_i , define

$$Y(x) := \{y \in C : h_i(x, y) \leq 0, i = 1, \dots, r\}, \quad v(x) := \min_{y \in Y(x)} g(x, y).$$

The following assumptions are in force throughout the main theorem.

Assumption 2.1 (Convex parametric lower problem). There is an open convex set $\Omega_0 \subset \mathbb{R}^n$ with $\Omega \subset \Omega_0$ such that:

- (A1) $C \subset \mathbb{R}^m$ is nonempty, compact, and convex.
- (A2) $g : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$ is finite, closed, and convex.
- (A3) For $i = 1, \dots, r$, $h_i : \Omega_0 \times \mathbb{R}^m \rightarrow \mathbb{R}$ is continuously differentiable and jointly convex.
- (A4) For every $\bar{x} \in \Omega$, there exist a neighborhood $U \Subset \Omega$ of $\bar{x}$, a point $\hat{y} \in C$, and a number $\sigma > 0$ such that

$$h_i(x, \hat{y}) \leq -\sigma, \quad x \in U, \quad i = 1, \dots, r.$$

Assumption 2.1(A4) implies in particular that $Y(x)$ is nonempty for all $x \in \Omega$. Since C is compact and g is continuous, the minimum defining $v(x)$ is attained.

Remark 2.2 (Why these assumptions are restrictive). The open problem motivating this note is broader than Assumption 2.1. In particular, the main theorem below does not cover arbitrary set-valued mappings $x \mapsto Y(x)$, nonconvex lower-level feasible regions, or general Lipschitz lower objectives. Joint convexity is used in the last step, where a limiting Lagrange multiplier formula is converted into an actual Clarke-subgradient inclusion. Compactness is used to avoid coercivity and level-boundedness issues.

3 The smoothing construction

For $\gamma > 0$, let $e_\gamma g$ denote the Moreau envelope

$$e_\gamma g(z) := \min_{w \in \mathbb{R}^{n+m}} \left\{ g(w) + \frac{1}{2\gamma} \|w - z\|^2 \right\}, \quad z = (x, y) \in \mathbb{R}^{n+m}.$$

Because g is finite closed convex, $e_\gamma g$ is convex and C^1 , with

$$\nabla e_\gamma g(z) = \frac{z - \text{prox}_{\gamma g}(z)}{\gamma} \in \partial g(\text{prox}_{\gamma g}(z)). \quad (1)$$

Set $p(t) := \frac{1}{2} [t]_+^2$. For $\mu > 0$, define

$$F_\mu(x, y) := e_\mu g(x, y) + \frac{1}{\mu} \sum_{i=1}^r p(h_i(x, y)) + \frac{\mu}{2} \|y\|^2, \quad (x, y) \in \Omega \times C, \quad (2)$$

that is,

$$F_\mu(x, y) = e_\mu g(x, y) + \frac{1}{2\mu} \sum_{i=1}^r [h_i(x, y)]_+^2 + \frac{\mu}{2} \|y\|^2.$$

The approximating value function is

$$v_\mu(x) := \min_{y \in C} F_\mu(x, y), \quad y_\mu(x) := \arg \min_{y \in C} F_\mu(x, y). \quad (3)$$

The minimizer is unique because $y \mapsto F_\mu(x, y)$ is μ -strongly convex on C : the envelope is convex, $p \circ h_i(x, \cdot)$ is convex because p is convex and nondecreasing while $h_i(x, \cdot)$ is convex, and $\frac{\mu}{2} \|y\|^2$ is strongly convex.

4 A Moreau-envelope lemma

We shall use the following elementary local form of Moreau approximation and gradient consistency.

Lemma 4.1 (Local Moreau estimates). *Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be finite closed convex, and let $K \subset \mathbb{R}^d$ be compact. Then there are constants $\mu_0 > 0$ and $L > 0$ such that, for every $z \in K$ and $0 < \mu \leq \mu_0$,*

$$g(z) - 2\mu L^2 \leq e_\mu g(z) \leq g(z), \quad \|\nabla e_\mu g(z)\| \leq 2L. \quad (4)$$

Moreover, if $z_k \rightarrow \bar{z}$, $\mu_k \downarrow 0$, and $\nabla e_{\mu_k} g(z_k) \rightarrow \eta$, then

$$\eta \in \partial g(\bar{z}). \quad (5)$$

Proof. Choose $\rho > 0$ and let $K_\rho := \{z : \text{dist}(z, K) \leq \rho\}$. Since g is finite convex, it is Lipschitz on K_ρ ; let L be a Lipschitz constant there. We first show that, after decreasing μ_0 if necessary, $\text{prox}_{\mu g}(z) \in K_\rho$ for all $z \in K$ and $0 < \mu \leq \mu_0$.

Indeed, convexity gives an affine minorant $g(w) \geq a + \langle q, w \rangle$ for some $a \in \mathbb{R}$ and $q \in \mathbb{R}^d$. Since g is bounded above on K and K is bounded, the inequality

$$g(w) + \frac{1}{2\mu} \|w - z\|^2 \leq g(z), \quad w = \text{prox}_{\mu g}(z),$$

implies

$$\frac{1}{2\mu} \|w - z\|^2 \leq A + B \|w - z\|$$

for constants A, B independent of $z \in K$ and small μ . Hence $\|w - z\| \rightarrow 0$ uniformly as $\mu \downarrow 0$, and therefore $w \in K_\rho$ for all sufficiently small μ .

For such μ , using $e_\mu g(z) \leq g(z)$ and the Lipschitz bound on K_ρ gives

$$\frac{1}{2\mu} \|w - z\|^2 \leq g(z) - g(w) \leq L \|w - z\|,$$

so $\|w - z\| \leq 2\mu L$. The gradient formula gives $\|\nabla e_\mu g(z)\| \leq 2L$. Also

$$e_\mu g(z) = g(w) + \frac{1}{2\mu} \|w - z\|^2 \geq g(z) - L \|w - z\| \geq g(z) - 2\mu L^2,$$

while $e_\mu g(z) \leq g(z)$ follows by testing $w = z$.

Finally, let $z_k \rightarrow \bar{z}$, $\mu_k \downarrow 0$, and set $w_k := \text{prox}_{\mu_k g}(z_k)$. Applying the preceding estimate on a compact set containing all large z_k shows $w_k - z_k \rightarrow 0$, hence $w_k \rightarrow \bar{z}$. By (1),

$$\nabla e_{\mu_k} g(z_k) = \frac{z_k - w_k}{\mu_k} \in \partial g(w_k).$$

The graph of the convex subdifferential is closed, so the limit η belongs to $\partial g(\bar{z})$. $\square$

5 Main theorem

Theorem 5.1 (Smoothing and gradient consistency under convex parametric constraints). *Suppose Assumption 2.1 holds, and define v_μ by (2)–(3). Then:*

(i) *For each fixed $\mu > 0$, $v_\mu \in C^1(\Omega)$. If*

$$\lambda_i^\mu(x) := \frac{[h_i(x, y_\mu(x))]_+}{\mu}, \quad i = 1, \dots, r,$$

then

$$\nabla v_\mu(x) = \nabla_x e_\mu g(x, y_\mu(x)) + \sum_{i=1}^r \lambda_i^\mu(x) \nabla_x h_i(x, y_\mu(x)). \quad (6)$$

(ii) *For every $\bar{x} \in \Omega$,*

$$\lim_{x \rightarrow \bar{x}, \mu \downarrow 0} v_\mu(x) = v(\bar{x}). \quad (7)$$

Thus $\{v_\mu\}_{\mu>0}$ is a smoothing family for v in the usual sense.

(iii) *The value function v is locally Lipschitz on Ω and, for every $\bar{x} \in \Omega$,*

$$\emptyset \neq \text{Limsup}_{x \rightarrow \bar{x}, \mu \downarrow 0} \nabla v_\mu(x) \subseteq \partial_C v(\bar{x}). \quad (8)$$

In fact, every cluster point ξ in the left-hand side satisfies the stronger local support inequality

$$v(x) \geq v(\bar{x}) + \langle \xi, x - \bar{x} \rangle \quad \text{for all } x \text{ near } \bar{x}, \quad (9)$$

so $\xi \in \widehat{\partial} v(\bar{x}) \subseteq \partial_C v(\bar{x})$.

5.1 Differentiability of the approximants

Proof of Theorem 5.1(i). For fixed $\mu > 0$, the function F_μ is continuous on $\Omega \times C$ and continuously differentiable in (x, y) on $\Omega \times \mathbb{R}^m$. As noted above, $F_\mu(x, \cdot)$ is strongly convex on C , and therefore $y_\mu(x)$ is unique. Berge's maximum theorem gives upper semicontinuity of the argmin mapping, and uniqueness upgrades this to continuity of $x \mapsto y_\mu(x)$.

Danskin's theorem then gives

$$\nabla v_\mu(x) = \nabla_x F_\mu(x, y_\mu(x)).$$

Since $p'(t) = [t]_+$, this is precisely (6). The right-hand side is continuous in x because $\nabla_x F_\mu$ and y_μ are continuous, so $v_\mu \in C^1(\Omega)$. $\square$

5.2 Approximation of the value function

Proof of Theorem 5.1(ii). Fix $\bar{x} \in \Omega$ and let $x_k \rightarrow \bar{x}$, $\mu_k \downarrow 0$. We prove $v_{\mu_k}(x_k) \rightarrow v(\bar{x})$. Let $y_k := y_{\mu_k}(x_k)$. Choose a Slater neighborhood $U \subseteq \Omega$ of $\bar{x}$, a Slater point $\hat{y} \in C$, and $\sigma > 0$ as in Assumption 2.1. For all large k , $x_k \in U$ and hence

$$F_{\mu_k}(x_k, y_k) \leq F_{\mu_k}(x_k, \hat{y}) = e_{\mu_k}g(x_k, \hat{y}) + \frac{\mu_k}{2} \|\hat{y}\|^2, \quad (10)$$

because all penalty terms vanish at $\hat{y}$. By Lemma 4.1, the right-hand side is bounded above, while $e_{\mu_k}g$ is bounded below on the compact set containing (x_k, C) for all large k . Thus

$$\sum_{i=1}^r [h_i(x_k, y_k)]_+^2 = O(\mu_k). \quad (11)$$

Passing to a subsequence if necessary, compactness of C gives $y_k \rightarrow \bar{y} \in C$. From (11) and the continuity of h_i , $h_i(\bar{x}, \bar{y}) \leq 0$ for every i , so $\bar{y} \in Y(\bar{x})$.

Using the nonnegativity of the penalty and regularization terms, Lemma 4.1, and the continuity of g ,

$$\liminf_{k \rightarrow \infty} v_{\mu_k}(x_k) = \liminf_{k \rightarrow \infty} F_{\mu_k}(x_k, y_k) \geq \liminf_{k \rightarrow \infty} e_{\mu_k}g(x_k, y_k) \geq g(\bar{x}, \bar{y}) \geq v(\bar{x}).$$

This proves the lower bound.

For the upper bound, fix any $y \in Y(\bar{x})$. Define

$$\varepsilon_k := \max_{1 \leq i \leq r} [h_i(x_k, y)]_+, \quad t_k := \frac{\varepsilon_k}{\sigma + \varepsilon_k}, \quad z_k := (1 - t_k)y + t_k \hat{y}.$$

Then $\varepsilon_k \rightarrow 0$, $t_k \rightarrow 0$, and $z_k \rightarrow y$. Since C is convex, $z_k \in C$. Since each $h_i(x_k, \cdot)$ is convex and $h_i(x_k, \hat{y}) \leq -\sigma$,

$$h_i(x_k, z_k) \leq (1 - t_k)h_i(x_k, y) + t_k h_i(x_k, \hat{y}) \leq (1 - t_k)\varepsilon_k - t_k \sigma = 0.$$

Thus $z_k \in Y(x_k)$. Therefore the penalty terms vanish at z_k , and

$$v_{\mu_k}(x_k) \leq F_{\mu_k}(x_k, z_k) = e_{\mu_k}g(x_k, z_k) + \frac{\mu_k}{2} \|z_k\|^2 \leq g(x_k, z_k) + \frac{\mu_k}{2} \|z_k\|^2 \rightarrow g(\bar{x}, y).$$

Taking the infimum over $y \in Y(\bar{x})$ gives $\limsup_k v_{\mu_k}(x_k) \leq v(\bar{x})$. Since the sequence (x_k, μ_k) was arbitrary, (7) follows. $\square$

5.3 Local Lipschitz continuity of the original value function

Lemma 5.2. *Under Assumption 2.1, v is locally Lipschitz on Ω .*

Proof. Fix $\bar{x} \in \Omega$ and choose a Slater neighborhood $U \Subset \Omega$ with point $\hat{y}$ and margin σ . Shrinking U if needed, assume it is convex and that g and all h_i are Lipschitz on a neighborhood of $\bar{U} \times C$. Let L_g be a Lipschitz constant for g there, let L_h be a common Lipschitz constant for the maps $x \mapsto h_i(x, y)$ on U uniformly over $y \in C$, and let $D := \text{diam}(C)$.

Take $x, x' \in U$ and $y \in Y(x)$. Set

$$\varepsilon := \max_i [h_i(x', y)]_+.$$

Since $y \in Y(x)$, $\varepsilon \leq L_h \|x' - x\|$. Define

$$t := \frac{\varepsilon}{\sigma + \varepsilon}, \quad z := (1 - t)y + t\hat{y}.$$

The same convexity calculation as above shows $z \in Y(x')$. Moreover,

$$\|z - y\| \leq Dt \leq \frac{D}{\sigma} \varepsilon \leq \frac{DL_h}{\sigma} \|x' - x\|.$$

Hence

$$g(x', z) \leq g(x, y) + L_g(\|x' - x\| + \|z - y\|) \leq g(x, y) + K \|x' - x\|,$$

where $K := L_g(1 + DL_h/\sigma)$. Taking $y \in \arg \min_{Y(x)} g(x, \cdot)$ gives $v(x') \leq v(x) + K \|x' - x\|$. Interchanging x and x' gives the reverse estimate, so v is Lipschitz on U . $\square$

5.4 Boundedness of penalty multipliers

Lemma 5.3 (Multiplier boundedness). *Fix $\bar{x} \in \Omega$. Let $x_k \rightarrow \bar{x}$, $\mu_k \downarrow 0$, and set*

$$y_k := y_{\mu_k}(x_k), \quad \lambda_{ik} := \frac{[h_i(x_k, y_k)]_+}{\mu_k}.$$

Then the sequence $\{\lambda_k\} \subset \mathbb{R}_+^r$ is bounded.

Proof. For all large k , work in a Slater neighborhood U of $\bar{x}$ with Slater point $\hat{y}$ and margin σ . The first-order optimality condition for minimizing $F_{\mu_k}(x_k, \cdot)$ over C is

$$\left\langle \nabla_y e_{\mu_k} g(x_k, y_k) + \sum_{i=1}^r \lambda_{ik} \nabla_y h_i(x_k, y_k) + \mu_k y_k, z - y_k \right\rangle \geq 0, \quad z \in C. \quad (12)$$

If $\lambda_{ik} > 0$, then $h_i(x_k, y_k) > 0$, and convexity of $h_i(x_k, \cdot)$ gives

$$\langle \nabla_y h_i(x_k, y_k), \hat{y} - y_k \rangle \leq h_i(x_k, \hat{y}) - h_i(x_k, y_k) \leq -\sigma.$$

Taking $z = \hat{y}$ in (12) therefore yields

$$0 \leq \langle \nabla_y e_{\mu_k} g(x_k, y_k) + \mu_k y_k, \hat{y} - y_k \rangle - \sigma \sum_{i=1}^r \lambda_{ik}.$$

Equivalently,

$$\sigma \sum_{i=1}^r \lambda_{ik} \leq \langle \nabla_y e_{\mu_k} g(x_k, y_k) + \mu_k y_k, \hat{y} - y_k \rangle.$$

The right-hand side is bounded above uniformly in k : C is compact, $\mu_k y_k$ is bounded, and Lemma 4.1 bounds $\nabla e_{\mu_k} g(x_k, y_k)$ on compact neighborhoods. Hence $\sum_i \lambda_{ik}$ is bounded. $\square$

5.5 Identification of cluster gradients

Proof of Theorem 5.1(iii). Local Lipschitz continuity was proved in Lemma 5.2. It remains to prove gradient consistency.

Let ξ be a cluster point of $\nabla v_{\mu_k}(x_k)$ for some $x_k \rightarrow \bar{x}$ and $\mu_k \downarrow 0$. Passing to a subsequence, we may suppose

$$\nabla v_{\mu_k}(x_k) \rightarrow \xi.$$

Set $y_k := y_{\mu_k}(x_k)$ and $\lambda_{ik} := [h_i(x_k, y_k)]_+ / \mu_k$. By compactness of C , Lemma 5.3, and Lemma 4.1, we may pass to a further subsequence such that

$$y_k \rightarrow \bar{y}, \quad \lambda_k \rightarrow \bar{\lambda} \in \mathbb{R}_+^r, \quad \nabla e_{\mu_k} g(x_k, y_k) \rightarrow \eta = (\eta_x, \eta_y).$$

By the approximation result already proved, $\bar{y} \in \arg \min_{y \in Y(\bar{x})} g(\bar{x}, y)$. Lemma 4.1 gives

$$\eta \in \partial g(\bar{x}, \bar{y}). \quad (13)$$

Furthermore, if $h_i(\bar{x}, \bar{y}) < 0$, then $h_i(x_k, y_k) < 0$ for all large k , and therefore $\lambda_{ik} = 0$ eventually. Hence

$$\bar{\lambda}_i h_i(\bar{x}, \bar{y}) = 0, \quad i = 1, \dots, r. \quad (14)$$

Passing to the limit in the variational inequality (12) gives

$$0 \in \eta_y + \sum_{i=1}^r \bar{\lambda}_i \nabla_y h_i(\bar{x}, \bar{y}) + N_C(\bar{y}). \quad (15)$$

Finally, passing to the limit in the gradient formula (6) yields

$$\xi = \eta_x + \sum_{i=1}^r \bar{\lambda}_i \nabla_x h_i(\bar{x}, \bar{y}). \quad (16)$$

We now prove that ξ supports v at $\bar{x}$. Let x be near $\bar{x}$ and let $y \in Y(x)$. Convexity of g and (13) imply

$$g(x, y) \geq g(\bar{x}, \bar{y}) + \langle \eta_x, x - \bar{x} \rangle + \langle \eta_y, y - \bar{y} \rangle. \quad (17)$$

Joint convexity and differentiability of h_i imply

$$h_i(x, y) \geq h_i(\bar{x}, \bar{y}) + \langle \nabla_x h_i(\bar{x}, \bar{y}), x - \bar{x} \rangle + \langle \nabla_y h_i(\bar{x}, \bar{y}), y - \bar{y} \rangle. \quad (18)$$

Multiplying (18) by $\bar{\lambda}_i$, summing in i , adding to (17), and using $g(\bar{x}, \bar{y}) = v(\bar{x})$, (14), and (16), we get

$$\begin{aligned} g(x, y) + \sum_{i=1}^r \bar{\lambda}_i h_i(x, y) &\geq v(\bar{x}) + \langle \xi, x - \bar{x} \rangle \\ &\quad + \left\langle \eta_y + \sum_{i=1}^r \bar{\lambda}_i \nabla_y h_i(\bar{x}, \bar{y}), y - \bar{y} \right\rangle. \end{aligned}$$

By (15), the last inner product is nonnegative for every $y \in C$. Since $y \in Y(x)$ and $\bar{\lambda} \geq 0$, we have $\sum_i \bar{\lambda}_i h_i(x, y) \leq 0$. Consequently,

$$g(x, y) \geq g(x, y) + \sum_{i=1}^r \bar{\lambda}_i h_i(x, y) \geq v(\bar{x}) + \langle \xi, x - \bar{x} \rangle.$$

Taking the infimum over $y \in Y(x)$ proves (9). Since v is locally Lipschitz, this implies $\xi \in \widehat{\partial}v(\bar{x}) \subseteq \partial_C v(\bar{x})$.

It remains only to note that the outer limit is nonempty. Fix any sequence $\mu_k \downarrow 0$ and take $x_k \equiv \bar{x}$. The preceding multiplier bound and Lemma 4.1 imply that $\{\nabla v_{\mu_k}(\bar{x})\}$ is bounded. Hence it has a convergent subsequence, whose limit belongs to $\text{Limsup}_{x \rightarrow \bar{x}, \mu \downarrow 0} \nabla v_\mu(x)$ by definition. This proves (8) and completes the proof of the theorem. $\square$

6 The smooth-objective quadratic-penalty special case

If g is already continuously differentiable, the Moreau envelope is not needed. This gives a closer version of the quadratic-regularization/penalization scheme mentioned in the open problem.

Corollary 6.1 (Pure quadratic penalty and vanishing regularization). *Suppose Assumption 2.1 holds, except that (A2) is strengthened to: g is continuously differentiable and jointly convex on a neighborhood of $\Omega \times C$. Define*

$$\tilde{F}_\mu(x, y) := g(x, y) + \frac{1}{2\mu} \sum_{i=1}^r [h_i(x, y)]_+^2 + \frac{\mu}{2} \|y\|^2, \quad \tilde{v}_\mu(x) := \min_{y \in C} \tilde{F}_\mu(x, y). \quad (19)$$

Then $\tilde{v}_\mu \in C^1(\Omega)$ for every $\mu > 0$,

$$\lim_{x \rightarrow \bar{x}, \mu \downarrow 0} \tilde{v}_\mu(x) = v(\bar{x}), \quad \bar{x} \in \Omega,$$

and

$$\emptyset \neq \text{Limsup}_{x \rightarrow \bar{x}, \mu \downarrow 0} \nabla \tilde{v}_\mu(x) \subseteq \partial_C v(\bar{x}), \quad \bar{x} \in \Omega.$$

If $\tilde{y}_\mu(x)$ denotes the unique minimizer in (19), then

$$\nabla \tilde{v}_\mu(x) = \nabla_x g(x, \tilde{y}_\mu(x)) + \sum_{i=1}^r \frac{[h_i(x, \tilde{y}_\mu(x))]_+}{\mu} \nabla_x h_i(x, \tilde{y}_\mu(x)).$$

Proof. Repeat the proof of Theorem 5.1 with $e_\mu g$ replaced by g . The differentiability and uniqueness arguments are unchanged. In the approximation proof, the estimates involving the Moreau envelope are replaced by continuity of g on compact neighborhoods. In the multiplier bound, $\nabla_y g(x_k, y_k)$ is bounded on compact neighborhoods. In the cluster-gradient argument, the limit vector is simply $\eta = \nabla g(\bar{x}, \bar{y})$, which belongs to $\partial g(\bar{x}, \bar{y})$ because g is convex and differentiable. The same Lagrangian support argument then proves the Clarke-subgradient inclusion. $\square$

7 What is and is not resolved

Theorem 5.1 and Corollary 6.1 establish a complete smoothing and gradient-consistency result for the stated convex inequality-described model with compact C and local uniform Slater feasibility. In this precise sense, they give a positive answer to the Alcantara–Takeda question under additional assumptions.

They should not be presented as a full solution of the general open problem. The proof relies essentially on:

- (i) compactness of C , or a replacement level-boundedness assumption strong enough to recover compactness of minimizing sequences;

- (ii) a local uniform Slater point, used both to repair feasible points and to bound penalty multipliers;
- (iii) convexity of the lower feasible graph, encoded here by joint convexity of the h_i and convexity of C ;
- (iv) joint convexity of g and h_i , used to turn limiting KKT data into a genuine supporting inequality for the value function.

The general x -dependent lower-level problem may fail any of these properties. Extending the result beyond this convex/Slater setting would require a different sensitivity argument, a different multiplier-boundedness mechanism, or a replacement for the convex Lagrangian support inequality.

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